

Improving Synchronization and Channel Estimation Robustness for OFDM-Based Integrated Sensing and Communication in GNU Radio

Akshita Muchhal
Project ID: g16_gsoc_isac (Individual Project)
Ahmedabad University

M1 Report – September 13, 2026

1 Problem Statement

1.1 Motivation

Integrated Sensing and Communication (ISAC) is an emerging wireless paradigm that permits a single transmitted waveform to support both data communication and radar-style sensing, thereby reducing spectrum and hardware duplication. OFDM is a natural carrier waveform for this purpose, as mature communication processing techniques (synchronization, channel estimation, equalization) already exist for it, and prior work has shown that OFDM radar processing (reciprocal filtering against known transmitted symbols) can extract range and Doppler information from a signal of the same form.

GNU Radio currently provides both components separately: OFDM transceiver blocks under `gr-digital`, and OFDM radar processing blocks under the `gr-radar` out-of-tree module. However, the existing codebase does not connect these two components for a single shared waveform.

1.2 Significance

The two processing chains were developed under different assumptions. Communication synchronization is designed to estimate and remove timing and frequency offsets so that data can be decoded. Radar sensing, by contrast, requires the propagation delay to be preserved, since it *is* the quantity being measured. Consequently, reusing the synchronization or channel-estimation output of one chain for the other may inadvertently remove information that the other function depends on.

Rather than treating synchronization and channel estimation as independently solved problems for each function, as expressed in

$$\text{minimize BER using } \hat{\tau}_{\text{comm}}, \hat{h}_{\text{comm}} \quad \text{and} \quad \text{minimize } \Delta R \text{ using } \hat{\tau}_{\text{radar}}, \hat{h}_{\text{radar}}, \quad (1)$$

this project investigates the consequences of requiring both functions to operate from a *shared* estimate of frame timing and channel state, $\hat{\tau}_{\text{shared}}, \hat{h}_{\text{shared}}$, obtained from the same received OFDM samples. The identification and resolution of this incompatibility constitutes a concrete, testable open-source contribution opportunity for GNU Radio, and is directly relevant to the broader 6G ISAC research agenda.

1.3 Assumptions

- A monostatic (or closely co-located) transmitter/receiver architecture, consistent with existing GNU Radio ISAC demonstrations on SDR hardware.
- A single OFDM numerology shared between the communication and sensing functions (no dual-numerology or dual-waveform design).
- Channel effects modeled as additive noise, multipath, and propagation delay, using GNU Radio’s channel modeling blocks in simulation; validation on real SDR hardware is treated as optional, time-permitting work.
- The specific open-source contribution (a bug fix, test-suite extension, or block modification) will be determined based on the findings of the M2 codebase and issue-tracker analysis, following GNU Radio’s established contribution process.

1.4 Scope

The scope of this project includes the following:

- Analysis of the `gr-digital` OFDM synchronization and channel-estimation blocks.
- Analysis of the `gr-radar` range and Doppler processing blocks.
- Identification of the points at which the assumptions of the two processing chains, regarding frame timing and channel state, diverge.
- Implementation of one concrete contribution that improves the robustness of a shared ISAC waveform.
- Quantitative evaluation via synchronization error, channel-estimation error, bit-error rate (BER), and range-estimation error across signal-to-noise ratio (SNR) and delay.

The following are outside the scope of this project, given the project timeline: multi-user or MIMO-ISAC extensions, standardized waveforms (e.g., 5G NR numerology), and hardware-in-the-loop validation beyond an optional SDR demonstration. Evaluation will primarily be conducted through simulation, with GNU Radio flowgraph results incorporated from M2 onward.

2 Review of Existing Work

2.1 OFDM Radar and ISAC Waveform Fusion

Sturm and Wiesbeck introduced reciprocal-filtering OFDM radar processing, in which received per-subcarrier symbols are divided by the known transmitted symbols and a two-dimensional IFFT/FFT is applied to obtain range and Doppler estimates. This constitutes the foundational technique on which both `gr-radar` and subsequent GNU Radio ISAC work are based; however, it assumes that communication-grade synchronization is already available and does not analyze synchronization or channel-estimation robustness.

2.2 ISAC System-Level Surveys and 6G Perspectives

Liu et al. survey joint radar-communication (JRC) designs (radar-centric, communication-centric, and jointly designed) together with the performance metrics used across the field (SINR, Cramér-Rao bound, spectral efficiency, ambiguity function). A subsequent paper by Liu et al. extends this survey into a 6G-focused tutorial covering waveform design and the sensing-communication performance trade-off region. Qaisar et al. address similar ground at a more network-oriented level, examining how ISAC fits into 6G architectures, associated enabling technologies (massive MIMO, RIS, mmWave/THz), and standardization efforts. All three sources indicate that synchronization

and channel-estimation robustness for a shared ISAC waveform remains an open problem, though none of them implement or evaluate a solution.

2.3 OFDM-Specific and AI-Based ISAC Signal Processing

Two additional recent papers are relevant to the signal-processing aspects of this project. Dai et al. present a detailed tutorial on MIMO-OFDM ISAC, including several frameworks for estimating delay, Doppler, and spatial parameters; this work is closely related to the radar side of the present problem, although it assumes that synchronization is already resolved rather than shared with an independent communication receiver. Vaezi et al. examine the application of deep learning to ISAC waveform and beamforming design, motivated by the trade-off between communication performance and sensing accuracy for a shared waveform; this represents a longer-term research direction rather than an approach anticipated for use in M2 or M3. Only the introductory and background sections of these two papers have been reviewed to date, and a more detailed reading will be undertaken alongside the M2 codebase analysis.

2.4 GNU Radio Realizations (Reference Work)

The most directly relevant prior work is that of Doly et al., who implemented and demonstrated a joint radar-communication waveform in GNU Radio using SDR hardware. Their synchronization approach is adapted from existing communication-oriented GNU Radio blocks, and the work constitutes primarily a functional demonstration rather than a systematic quantitative evaluation. The `gr-radar` module itself implements OFDM radar processing as reusable GNU Radio blocks but assumes that frame timing is already established and is not designed to interoperate with `gr-digital`'s synchronization or channel-estimation blocks. This constitutes the specific integration gap addressed by this project.

3 Technical Progress and SOTA Positioning

3.1 Technical Progress

The majority of the effort during M1 was directed toward literature review rather than the implementation of GNU Radio code. This review examined how `gr-digital` performs OFDM synchronization and channel estimation, and how `gr-radar` recovers range and Doppler through reciprocal filtering, in conjunction with the papers discussed above. This review indicated that the two modules are developed independently and do not currently exchange synchronization or channel information, which forms the starting point for the M2 codebase analysis.

3.2 SOTA Positioning

Table 1 summarizes how the reviewed works relate to the specific problem addressed by this project.

3.3 Research Gap

Each reviewed work addresses one side of the problem: radar-style estimation theory (Sturm & Wiesbeck), system-level ISAC surveys (Liu et al., Qaisar et al.), OFDM-specific parameter estimation (Dai et al.), or an actual GNU Radio implementation (Doly et al., `gr-radar`). However, none of these works quantify the specific coupling addressed by this project: the effect on communication

Work	OFDM Waveform	GNU Radio Impl.	Sync/CFO Robustness	Radar Range/Channel Est.	Joint Comm+Radar Eval.
Sturm & Wiesbeck (2011)	Yes	No	No	Yes	No
Liu et al. (2020)	Survey	No	No	Survey	No
Liu et al. (2022)	Survey	No	No	Survey	No
Dai et al. (2025)	Yes	No	No	Yes	Partial
Vaezi et al. (2026)	Partial	No	No	Partial	Partial
Qaisar et al. (2026)	No	No	No	No	No
Doly et al. (2023)	Yes	Yes	No	Yes	Partial
KIT-CEL, <code>gr-radar</code>	Yes	Yes	No	Yes	No
Proposed Project	Yes	Yes	Yes	Yes	Yes

Table 1: Comparison of related work and the proposed project. “Sync/CFO Robustness” denotes whether the work analyzes or quantifies the effect of imperfect synchronization on the other function; “Joint Comm+Radar Quantitative Eval.” denotes whether BER/channel-estimation and range/Doppler error are measured together on the same waveform.

and radar performance *jointly* when both are required to share a single frame-timing and channel estimate on the same OFDM waveform in GNU Radio. This project addresses that gap directly.

3.4 Proposed Methodology

The proposed work will adopt the general `gr-digital/gr-radar` architecture already available in GNU Radio. In M2, the relevant source code and issue tracker will be analyzed in detail to identify a specific, scoped contribution (a bug fix, test-suite extension, or block modification). In M3, this contribution will be implemented in a GNU Radio flowgraph and tested in accordance with GNU Radio’s contribution guidelines. In M4, the effect of the contribution will be evaluated quantitatively using synchronization error, channel-estimation error (NMSE), BER, and range-estimation error, swept across SNR and target delay, and compared against the current (unmodified) behavior as a baseline. Results will also be compared, where relevant, against the SOTA reference (Doly et al.).

4 Timeline

5 References

1. C. Sturm and W. Wiesbeck, “Waveform design and signal processing aspects for fusion of wireless communications and radar sensing,” *Proceedings of the IEEE*, vol. 99, no. 7, pp. 1236–1259, 2011.
2. F. Liu, C. Masouros, A. P. Petropulu, H. Griffiths, and L. Hanzo, “Joint radar and communication design: Applications, state-of-the-art, and the road ahead,” *IEEE Trans. Commun.*, vol. 68, no. 6, pp. 3834–3862, 2020.
3. F. Liu *et al.*, “Integrated sensing and communications: Toward dual-functional wireless networks for 6G and beyond,” *IEEE J. Sel. Areas Commun.*, vol. 40, no. 6, pp. 1728–1767, 2022.
4. S. A. Doly, A. Chiriyath, W. Standage-Beier, and D. W. Bliss, “GNU Radio realization of Waveform Co-design for Joint Radar-Communications system using SDRs,” *Proceedings of the GNU Radio Conference*, vol. 8, no. 1, 2023.
5. Communications Engineering Lab, Karlsruhe Institute of Technology, “gr-radar,” GitHub repository. [Online]. Available: <https://github.com/kit-cel/gr-radar>

Stage	Proposed dates	Planned Work
M1	Sep 13 – Sep 27, 2026	Problem definition, literature review, and SOTA positioning.
M2	Sep 28 – Oct 11, 2026	Detailed analysis of gr-digital and gr-radar source code and issue tracker; identification of a specific, scoped contribution; initial discussion with maintainers/community.
M3	Oct 12 – Oct 25, 2026	Implementation of the contribution in a GNU Radio flowgraph; unit and integration testing in accordance with GNU Radio contribution guidelines.
M4	Oct 26 – Nov 8, 2026	Quantitative evaluation (synchronization error, channel-estimation error, BER, range error) across SNR and delay; comparison against the current GNU Radio behavior and the SOTA reference.
M5	Nov 9 – Nov 30, 2026	Result analysis, documentation, GitHub repository finalization, pull request submission (if accepted by maintainers), and preparation of the final report and video.

Table 2: Proposed milestone timeline (September–November 2026). Dates will be adjusted once official milestone deadlines are confirmed via Campuswire.

6. Q. Dai, Y. Zeng, H. Wang, C. You, C. Zhou, H. Cheng, X. Xu, S. Jin, A. L. Swindlehurst, Y. C. Eldar, R. Schober, R. Zhang, and X. You, “A Tutorial on MIMO-OFDM ISAC: From Far-Field to Near-Field,” *arXiv preprint*, arXiv:2504.19091, 2025.
7. M. Vaezi, G. Amarasuriya Aruma Baduge, E. Ollila, and S. A. Vorobyov, “A Tutorial on AI-Empowered Integrated Sensing and Communications,” *IEEE Communications Surveys & Tutorials*, vol. 28, 2026.
8. M. U. F. Qaisar, W. Yuan, O. Günlü, T. Riihonen, Y. Cui, N. Gonzalez-Prelcic, L. Zhang, M. Di Renzo, and Z. Han, “The Role of ISAC in 6G Networks: Enabling Next-Generation Wireless Systems,” *IEEE Trans. Network Science and Engineering*, vol. 13, 2026.